Beyond the Verbal Loop: Graph Traversal, Manifold Cognition, Self-Sealing Knots, and the Architecture of Representational Freedom

A Monograph in Cognitive Epistemology, Formal Logic, and Contemplative Science

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Abstract

"There are no absolute truths to be found in words alone." Far from being merely a contemplative aphorism, this proposition reflects a structural constraint of symbolic representation: while formal linguistic systems derive deductively valid consequences from explicit axioms, classic limitative theorems in mathematical logic and computability theory—most notably Gödel's incompleteness theorems [1], Tarski's undefinability of truth [2], and Rice's theorem on the undecidability of non-trivial semantic properties [4, 5]—demonstrate that sufficiently expressive symbolic systems cannot rely solely on internal syntactic manipulation to adjudicate their own global truth or semantic grounding. Motivated by this formal boundary, this monograph develops an integrated theoretical and empirical framework examining how serial verbal reasoning generates self-referential cognitive traps (*verbal knots*) and how multidimensional and metacognitive processes enable cognitive systems to escape them. Maintaining a strict separation among (1) **proposed theoretical constructs**, (2) **established mathematical, psychological, and philosophical results**, and (3) **preregisterable empirical hypotheses**, the analysis first contrasts step-by-step symbolic *graph traversal* with multidimensional *manifold cognition* [36, 38, 39]. It then formalizes four structural mechanisms of verbal loop traps—*semantic drift*, *axiomatic stacking*, *reflexive inoculation* (*epistemic black holes* [6]), and *inferential ratcheting*—and applies them to long-form deductive corpora such as existential-risk ("doomer") argumentation [9]. Finally, the monograph examines the *meta-knot* of the self-model [18, 23], qualifies the role of intelligence in rationalization [11, 15], contrasts self-perpetuating *selfish knots* with paradoxical *selfless knots* (kōans) [44, 46], and outlines a preregisterable experimental programme distinguishing *medium-relative freedom* (*representation change*) from *representational freedom* (*epistemic revision* and *decentering* [49]).

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1 Formal Limits of Symbolic Systems and the Verbal Loop Trap

There are no absolute truths to be found in words alone. This proposition is more than a philosophical or contemplative observation: natural and formal languages operate as rule-governed symbolic calculi, and whenever a symbolic system achieves sufficient expressive power to represent its own statements, it encounters rigorous mathematical boundaries established by Kurt Gödel, Alfred Tarski, and Henry Gordon Rice [1, 2, 4, 5]. Understanding what these limitative theorems formally prove—and how they motivate a cognitive theory of self-referential delusion—provides the foundation for analyzing how human minds become trapped in, and liberated from, verbal knots.

1.1 Gödel, Tarski, and Rice: What Formal Logic Proves vs. What Cognition Analogue-Models

In mathematical logic and theoretical computer science, three landmark results define the intrinsic horizon of purely symbolic manipulation:

1. **Gödel's First and Second Incompleteness Theorems (1931):** Any consistent, effectively axiomatized formal system $\mathcal{F}$ capable of expressing elementary arithmetic contains well-formed statements $G_{\mathcal{F}}$ that are true in the standard model yet unprovable within $\mathcal{F}$, and $\mathcal{F}$ cannot prove its own consistency ($\mathrm{Con}(\mathcal{F})$) using only its internal deductive apparatus [1, 4].
2. **Tarski's Undefinability Theorem (1936):** No sufficiently rich formal language $\mathcal{L}$ can define its own truth predicate $\mathrm{True}_{\mathcal{L}}(\ulcorner\phi\urcorner)$ without generating semantic paradoxes (such as the Liar cycle $\lambda \leftrightarrow \neg\mathrm{True}_{\mathcal{L}}(\ulcorner\lambda\urcorner)$). Consequently, the semantic truth of propositions in $\mathcal{L}$ can only be adjudicated within a strictly richer meta-language $\mathcal{L}'$ [2, 3, 4].
3. **Rice's Theorem (1953):** In computability theory, for any non-trivial semantic (extensional) property $\mathcal{P}$ of partial computable functions, there exists no general algorithmic procedure that can inspect an arbitrary symbolic program description e and decide whether the function computed by e satisfies $\mathcal{P}$ [4, 5]. Recent extensions of Rice's theorem to self-modifying and self-referential computational agents confirm that no internal syntactic verifier can globally decide the semantic soundness or safety of its own recursive updates without an external evaluator or oracle [5, 4].

Methodological rigor requires drawing a sharp boundary between these formal theorems and their application to human psychology. Natural language is not a rigid first-order Peano arithmetic, nor is human thought a Turing machine executing static code. Accordingly, Gödel's, Tarski's, and Rice's theorems do **not** establish that words cannot express truth; rather, they prove a precise structural principle: **internal syntactic coherence cannot guarantee external semantic grounding** [4]. A symbolic system can be locally valid at every deductive step while remaining globally ungrounded, circular, or undecidable from within its own vocabulary.

1.2 The Cognitive Hypothesis: "Wordcel" Loop Traps vs. "Shape Rotator" Inspection

Motivated by this structural boundary, we formulate a cognitive distinction between two functional modes of representation—colloquially contrasted in contemporary cognitive typology as the *verbalizer* ("wordcel") mode and the *visuospatial/multidimensional* ("shape rotator") mode [36]:

- **The Serial Symbolic Mode ("Wordcel" Graph Traversal):** When cognition relies exclusively on sequential verbal propositions, inference proceeds step-by-step along chains of lexical definitions. Because words are arbitrary tokens decoupled from physical geometry [36], a purely verbal reasoner can construct elaborate, internally consistent deductive loops—closed cycles in a conceptual graph, self-referential fixed points, or definitions that subtly presuppose their own conclusions—while experiencing the internal fluency of the chain as proof of objective truth.
- **The Multidimensional Mode ("Shape Rotator" Manifold Inspection):** By contrast, when a problem is embedded into a non-verbal, continuous multidimensional representation (such as a spatial simulation, geometric manifold, or cross-modal relational array [36, 37]), the cognitive system acts analogously to Tarski's meta-language $\mathcal{L}'$: it steps outside the 1D symbolic chain and evaluates the entire relational structure simultaneously against orthogonal spatial, physical, or counterfactual constraints. What appeared inside the verbal chain as an inescapable deductive necessity is directly seen from the higher-dimensional vantage point as a closed, self-referential loop.

Core Epistemic Thesis: A cognitive system can gain the ability to represent its own constraints without necessarily becoming free of them. Cognitive liberation requires not only representational flexibility—shifting from serial verbal chains to multidimensional models—but a changed relationship to the representations through which the system models itself and the world.

2 Graph Traversal versus Manifold Cognition: Topology of Verbal Loop Traps

To analyze how intelligent reasoners become entrapped in purely verbal arguments, we can model the structural difference between serial propositional reasoning and multidimensional cognition using the contrast between **discrete graph traversal** and **continuous manifold representation**.

2.1 The Graph Model vs. The Manifold Model

In a **discrete conceptual graph** $G = (V, E)$, nodes $v_i \in V$ represent discrete lexical concepts or propositional claims, and directed edges $e_{ij} \in E$ represent local licensed inferences ($v_i \implies v_j$). A serial verbal reasoner traverses G locally, holding only a narrow window of adjacent nodes (v_{t-1}, v_t, v_{t+1}) in working memory at any instant [36]. If the graph contains a long directed cycle $v_1 \to v_2 \to \cdots \to v_k \to v_1$ (where k exceeds the span of working memory), the local traveler experiences an unending, forward-moving deductive corridor without recognizing that the trajectory is walking in a circle.

In a **continuous multidimensional manifold** $\mathcal{M}$, by contrast, states are represented across multiple simultaneous axes of variation, permitting global inspection of local curvature, boundary conditions, gradients, and topological holes [36, 38]. Using algebraic topology as a **heuristic structural analogy** (rather than a literal physical identity), a self-sealing verbal loop behaves analogously to a non-contractible closed loop $\gamma : [0, 1] \to \mathcal{X}$ around a puncture or obstruction in a space with a non-trivial fundamental group $\pi_1(\mathcal{X}) \neq 0$ [29]: so long as the thinker remains confined to the 1D path metric of the punctured plane, the loop γ cannot be continuously contracted to a point. Only when the representation is embedded into a space that fills in the omitted variable or adds an orthogonal dimension can the loop be lifted and untied.

2.2 Four Structural Mechanisms of Verbal Loop Traps

When an argument or belief system entraps a serial reasoner, it typically relies on four specific structural mechanisms. Table 1 formalizes each mechanism both at the level of local propositional rhetoric and at the level of global conceptual topology.

2.3 Epistemic Black Holes and the Decoupling of Certainty from Support

The third mechanism in Table 1—**reflexive inoculation**—corresponds directly to what Maarten Boudry and Steije Hofhuis (2024) formalize in *Theoria* as **epistemic black holes**: self-sealing belief systems (such as conspiracy theories or closed ideologies) that evolve built-in immunization clauses (`Ideology` $\cap$ `immune_response`) [6, 7]. Expanding on Karl Popper's analysis of reinforced dogmatism and Imre Lakatos's concept of protective auxiliary belts [6], Boudry and Hofhuis show that once a believer crosses the "event horizon" of a self-sealing system, any counterevidence $\neg E$ is automatically explained away as a cover-up, fabrication, or symptom of the critic's delusion.

In formalizing this mechanism as a **selfish knot** ($\mathcal{K}_{\text{selfish}}$), it is essential to avoid a common Bayesian fallacy. Suppose a core hypothesis H is conjoined with an immunizing auxiliary I asserting that hostile forces systematically fabricate counterevidence $\neg E$ whenever H is true. Within the believer's **subjective likelihood model**, observing $\neg E$ yields a subjective likelihood ratio greater than unity:

$$\Lambda_{\text{subj}}(\neg E) = \frac{P_{\text{subj}}(\neg E \mid H \wedge I)}{P_{\text{subj}}(\neg E \mid \neg(H \wedge I))} > 1 \quad \text{(conditional on auxiliary } I\text{)} \tag{1}$$

Crucially, Inequality (1) does **not** imply that every counterexample legitimately increases the objective evidential support for H. First, the inequality holds only under the agent's internal, unverified assumption I. Second, as Boudry and Hofhuis emphasize, self-sealing immunization generates **arbitrariness**

Table 1: The Four Structural Mechanisms of Verbal Loop Traps: Propositional vs. Topological Dynamics

Loop Mechanism	**Propositional / Rhetorical Operation (1D Graph View)**	**Topological Interpretation (Manifold View)**	**Diagnostic Failure Mode**
1. Semantic Drift (Definition Shifting)	Subtle, unannounced shifts in the intensional definition of a key term across a multi-step argument chain ($T_1 \approx T_2 \approx \cdots \not\approx T_n$).	Continuous holonomy or parallel-transport twist across conceptual space, returning to the starting label at a displaced semantic coordinate.	Equivocation masked by lexical continuity; premises true under sense A used to prove conclusions under sense B.
2. Axiomatic Stacking	Layering multiple "modest" or locally plausible premises ($A_1 \wedge A_2 \wedge \cdots \wedge A_m$) until the composite assumption becomes extremely brittle yet treated as bedrock.	Imposing successive boundary hyperplanes that narrow the allowable state space into a measure-zero corridor while hiding the external volume.	High joint fragility $\prod_{i=1}^{m} P(A_i) \ll 1$ disguised as step-by-step common sense.
3. Reflexive Inoculation (Epistemic Black Holes)	Framing any counterargument, empirical absence, or skepticism as diagnostic evidence of the critic's bias, complicity, or cognitive failure ([6]).	A topological attractor basin with inward-pointing vector fields at its boundary, routing external perturbation trajectories back into the loop.	Unfalsifiable self-sealing closure; subjective certainty decouples from predictive validity ([6]).
4. Inferential Ratcheting	Asymmetric, unidirectional inferential transitions where tentative possibilities at step t are cited as established lemmas at step $t+1$.	A directed graph with diode-like edges permitting forward traversal toward an extreme attractor while blocking backward audit of priors.	Runaway extremity; conclusions carry far greater certainty than the product of their inferential links warrants.

and explanatory proliferation [6]: because the auxiliary belt I is elastic enough to accommodate *any* observation (E or $\neg E$), the composite model $H \wedge I$ loses discriminative power and suffers a severe Occam marginal-likelihood penalty. Thus, the defining pathology of a *selfish knot* is that **it preserves or inflates subjective certainty while its predictive usefulness and correspondence with reality deteriorate** [6].

2.4 Case Study in Deductive Graph Closure: Existential-Risk ("Doomer") Discourse

How do these four mechanisms interact in high-intelligence, non-conspiratorial intellectual communities? A prominent contemporary domain where critics and analysts have debated the tension between serial verbal deduction and multidimensional world-modeling is **AI existential-risk ("doomer") discourse**, exemplified by long-form deductive corpora such as Eliezer Yudkowsky's *Sequences* and related arguments on LessWrong [8, 9].

Yudkowsky's corpus is notable for its explicit commitment to Bayesian epistemology, decision theory, and dense chain-of-thought deductive argumentation [8]. Yet critics within the technical and philosophical community have argued that certain unconditional "doom" trajectories exhibit the structural vulnerability of 1D graph traversal over multidimensional manifold grounding [9]:

- *Axiomatic Stacking and Corridor Narrowing:* Plausible individual premises regarding instrumental convergence, discontinuity of capability gain, and deceptive alignment are stacked sequentially; while each link is argued forcefully in prose, the joint probability across the continuous manifold of real-world engineering, feedback, and incremental deployment constraints is rarely integrated globally [9].
- *Semantic Drift and Inferential Ratcheting:* Terms such as "optimization," "agency," or "alignment"

can shift between narrow technical definitions and broad anthropomorphic or teleological senses across long essays, ratcheting conditional theoretical possibilities into near-certainty.

- *Reflexive Inoculation:* Skepticism toward apocalyptic timelines is sometimes diagnosed within the framework itself as a predictable cognitive bias (e.g., scope insensitivity, status-quo bias, or inability to visualize exponential curves), insulating the deductive chain from external calibration.

When such a verbal knot forms in any intellectual community, **three distinct generative possibilities** must be distinguished:

1. **Deliberate Rhetorical Construction:** An author knowingly engineers definitions and ratchets to compel agreement while remaining privately aware of the argument's boundary failures.
2. **Motivated Construction:** An author unconsciously selects premises and auxiliary defenses to protect an affectively or socially invested conclusion [11, 13].
3. **Shared Representational Capture:** Both author and audience communicate with complete sincerity, but because both operate exclusively within the same constrained 1D symbolic graph, both mistake internal deductive coherence for objective correspondence with reality.

Establishing whether a specific corpus (such as Yudkowsky's *Sequences* [9] or any other foundational text) contains genuine rhetorical knots rather than valid warnings cannot be settled by spatial metaphors alone; it requires a three-tier empirical audit: (a) **Local Logic Analysis** (testing individual premise validity), (b) **Argument Topology Reconstruction** (mapping directed dependency graphs to detect hidden cycles, definition shifts, and omitted manifold dimensions), and (c) **Epistemic Behavior Testing** (specifying concrete falsifiers and testing whether conclusions survive translation into non-verbal, quantitative, or simulation-based representations).

3 The Architecture of the Meta-Knot and the Intelligence Paradox

Why are highly intelligent, analytically trained thinkers susceptible to *shared representational capture* and self-sealing knots? Answering this question requires shifting focus from the structure of external arguments to the internal architecture of the **self-model** that evaluates them.

3.1 The Self as Meta-Knot: Strange Loops and Phenomenal Transparency

Our central theoretical construct is that **the self-model is the primary meta-knot**: a recursive representational loop that participates in, and defends, the subsidiary cognitive knots it attempts to inspect. Three established frameworks in cybernetics and philosophy of mind ground this construct:

1. **Strange Loops and Cybernetic Self-Reference:** In Douglas Hofstadter's formulation, the "I" emerges as a **strange loop** within a tangled symbolic hierarchy, where top-level summary representations feed back downward to gate lower-level perception and inference [23]. In George Spencer-Brown's *Laws of Form* and Louis Kauffman's *Knot Logic*, such self-referential re-entry ($J = \overline{J}$) generates persistent oscillatory structure [29, 30, 31].
2. **Laing's Interpersonal Knots and Bateson's Double Bind:** R.D. Laing's *Knots* (1970) [24, 25] and Gregory Bateson's cybernetic theory of the **double bind** [26, 27] demonstrate that psychological entanglement requires a primary binding rule paired with a secondary meta-rule (at Logical Type $n + 1$) that prohibits recognizing or stepping outside the frame.
3. **Metzinger's Phenomenal Transparency:** Thomas Metzinger's **Self-Model Theory of Subjectivity (SMT)** explains why a cognitive system experiences its knots "from inside the string" [18, 20, 19]. The brain constructs a dynamic **Phenomenal Self-Model (PSM)** [21, 22].

Table 2: Distinguishing Proposed Model Parameters from Established Literature and Empirical Proxies

Proposed Parameter	**Epistemic Status**	**Theoretical Role in Meta-Knot Model**	**Operational Proxies & Falsification Criteria**
Transparency Weight ($\tau \in [0,1]$)	Proposed Model Variable (inspired by Metzinger [18]).	Indexes the degree of cognitive fusion ($\tau \to 1$) vs. metacognitive recognition of representations as models ($\tau \to 0$).	Inverse scores on the Experiences Questionnaire (EQ) Decentering scale [49, 52]; State Cognitive Defusion assays [53].
Apex Self-Prior Precision (Π_L)	Proposed Model Variable (extending Laukkonen & Slagter [46]).	Precision weight assigned to temporally deep identity priors relative to sensory/empirical prediction error (Π_1).	Hierarchical Gaussian Filter (HGF) prior-rigidity estimates during identity-salient belief updating; error-related negativity (ERN).
Precision Release ($\Delta\Pi_L < 0$)	Testable Mechanism for Selfless Knots / Decentering.	Hypothesized relaxation of top-down narrative rigidity following paradoxical exhaustion or disidentification training.	Increased post-intervention Bayesian belief revision paired with reduced default-mode network (DMN) narrative coupling [48].

Because the sub-personal neural processes generating the PSM are invisible to introspection, the self-model is **phenomenally transparent**: the organism looks *through* the representation directly at its content, mistaking a constructed internal model for an irreducible ontological entity [18, 19].

3.2 Candidate Neurocomputational Formalization (Proposed Model Variables)

Extending Metzinger's philosophical concept of transparency [18] and Ruben Laukkonen and Heleen Slagter's (2021) theoretical model of meditation within predictive processing [46, 48], we propose a formal parameterization of the Meta-Knot. We explicitly note that the variables below are **proposed theoretical parameters** requiring empirical operationalization (Table 2), not established neurobiological measurements.

Definition 3.1 (Proposed Meta-Knot and Schematic Precision Equations)**.** Let $S_k(\mathcal{M}_{\text{PSM}})$ denote the apex self-referential state in a recursive hierarchy $\mathcal{K}_{\text{self}} : S_0 \to S_1 \to \cdots \to S_k \to S_1$, and let $\tau \in [0,1]$ denote a proposed **transparency parameter** ($\tau \to 1$ indicating complete phenomenal transparency/fusion; $\tau \to 0$ indicating phenomenal opacity, where the representation is recognized as a constructed model). Under a simplified, schematic Gaussian–Laplace approximation assuming conditional independence across L cortical hierarchy levels, the precision-weighted prediction error objective is written schematically as:

$$\mathcal{F}_{\text{schematic}} \approx \sum_{l=1}^{L} \Pi_l \, \|\varepsilon_l\|^2 = \sum_{l=1}^{L} \frac{1}{\sigma_l^2} \left(\mu_l - g_l(\mu_{l+1})\right)^2 \tag{2}$$

where μ_l is the state expectation at level l, $g_l(\mu_{l+1})$ is the descending prediction, and $\Pi_l = \sigma_l^{-2}$ is the **precision weight**. In our proposed model, the **Meta-Knot** ($\mathcal{K}^*$) corresponds to the regime where $\tau \to 1$ and the temporally deep narrative self-prior μ_L is assigned disproportionately high precision ($\Pi_L \gg \Pi_1$), causing the system to recruit auxiliary top-down inferences g_l to explain away disconfirming bottom-up prediction errors ε_l whenever a belief fused with μ_L is challenged.

As Laukkonen and Slagter (2021) emphasize, predictive-processing models of self-deconstruction remain theoretical proposals that call for rigorous neurophenomenological validation [46, 48]. Table 2 defines the empirical proxies required to test and falsify τ and Π_L.

3.3 The Intelligence Paradox and the "Projection vs. Liberation" Trap

When an individual with high fluid intelligence (g), verbal fluency, or quantitative numeracy confronts a cognitive knot, a subtle trap emerges: **projection is not liberation**. A sophisticated thinker can

step up from Logical Type n to Logical Type $n+1$, projecting the knot into a higher-dimensional analytical or meta-theoretical model—describing, categorizing, and diagramming their own biases from all angles—without actually untying the underlying affective and egoic loop ($\tau \to 1$).

What does empirical psychology establish regarding the relationship between intelligence and self-delusion?

- **Myside Bias Independence from Intelligence (Stanovich, West, & Toplak):** In their comprehensive review *"Myside Bias, Rational Thinking, and Intelligence"* [11, 12] and *What Intelligence Tests Miss* [10], Keith Stanovich, Richard West, and Maggie Toplak demonstrate that **the magnitude of myside bias shows virtually no correlation with general intelligence**. This establishes that high cognitive ability does *not* automatically protect a thinker against biased belief maintenance [11]. However, it does not by itself prove that intelligence universally increases delusion.
- **Mixed Replications of Motivated Numeracy (Kahan et al. vs. Persson et al.):** While Kahan et al. (2017) reported that higher numeracy was associated with greater ideological polarization on a contingency-table task [13, 14], a large preregistered replication by Emil Persson and colleagues (2021) in *Cognition* **did not find robust evidence that motivated reasoning reliably increases with numeracy**, concluding that the cumulative evidence is mixed [15, 16, 17].
- **Functional Cognitive Modes Without Hemispheric Mythology:** Similarly, while Iain McGilchrist's *The Master and His Emissary* [34, 35] offers a rich philosophical allegory of a closed, certainty-seeking "emissary" usurping a holistic "master," reviews by Michael Corballis (*"Left Brain, Right Brain: Facts and Fantasies"* [32]) and neurophilosophical critiques of the Divided Brain Hypothesis [33] caution against mapping closed serial logic and open metaphorical understanding onto literal left and right cerebral hemispheres. Our framework models these as distributed **functional cognitive regimes**, not hemispheric homunculi.

Integrating these findings yields a precise, testable hypothesis rather than an overgeneralized claim:

Empirical Hypothesis 3.1 (Conditional Rationalization and Meta-Projection Hypothesis)**.** Higher cognitive ability (g, verbal fluency, and numeracy) may amplify the capacity to construct and defend complex auxiliary rationalizations—and to build $n+1$-dimensional meta-models that simulate insight without unbinding the self-loop—when reasoning is recruited in the service of an identity-protective belief. Whether this capacity increases the persistence or subjective certainty of self-sealing beliefs depends on moderating cognitive and motivational conditions, specifically the elasticity of available auxiliary hypotheses [6] and the degree of metacognitive fusion (τ) with the self-model.

4 Heuristic Representational Regimes, Selfless Knots, and Representational Freedom

If 1D verbal graph traversal is vulnerable to closed loops and $n+1$ analytical projection can merely relocate a knot, how does a cognitive system achieve genuine liberation? We organize the pathways of cognitive release across a **heuristic hierarchy of representational regimes** (Table 3) and contrast two distinct mechanisms: (1) intra-symbolic paradoxical exhaustion via **selfless knots (kōans)**, and (2) cross-modal transformation via **wordless visualization and disidentification**.

4.1 A Heuristic Hierarchy of Representational Regimes

Table 3 orders five representational regimes from serial verbal chains (1D) through poetic metaphor (1.5D), schematic visual diagrams (2D), and dynamic spatial world-models (3D–4D) to metacognitive opacity ($\tau \to 0$).

Table 3: Heuristic Hierarchy of Representational Regimes and Their Proposed Functions

Representational Regime	**Heuristic Index**	**Proposed Cognitive Function**	**Established Cognitive Science Correlate**
1. Linear Verbal Sequence	1D (Heuristic)	Serial symbolic inference along discrete lexical tokens and directed graph edges.	Verbal *logogen* subsystem in Dual-Coding Theory ([36]); phonological loop.
2. Poetic and Metaphorical Language	1.5D (Heuristic)	Polysemous associations and cross-domain structural mappings that bypass rigid single-domain closure.	Conceptual Metaphor Theory ([38]) and Fauconnier–Turner Conceptual Blending ([39, 40]).
3. Schematic Visual Representation	2D (Heuristic)	Simultaneous planar inspection of relational topology, causal cycles, and network structure.	Diagrammatic reasoning and visuospatial sketchpad arrays ([36, 37]).
4. Dynamic World Model	3D–4D (Heuristic)	Parallel integration of volumetric spatial, temporal, and counterfactual state transitions (manifold cognition).	Non-verbal *imagen* mental simulation ([36]) and generative world-models ([48]).
5. Metacognitive or Contemplative Awareness	Opacity ($\tau \to 0$)	Recognition of the constructed status of *all* representations across verbal and visual media (*Representational Freedom*).	Decentering ([49, 51]), Cognitive Defusion ([53]), and Phenomenal Opacity ([18]).

Crucially, the numerical values 1D, 1.5D, 2D, and 3D–4D in Table 3 are **explicitly heuristic indices**, not established metric or Hausdorff fractal measurements in cognitive science. Similarly, while geometric topology proves that any 1-knot embedded in $\mathbb{R}^3$ ($S^1 \hookrightarrow \mathbb{R}^3$) can be smoothly untied when lifted into $\mathbb{R}^4$ [29], we retain this result strictly as a **mathematical analogy** motivating the hypothesis that expanding relational concurrency—via Conceptual Blending [39, 41, 42, 43] or Paivio's synchronous *imagen* system [36]—helps reasoners detect and bypass circularities hidden in serial prose.

4.2 Selfish Knots vs. Selfless Knots: The Paradox of the Kōan

How can a **Zen kōan**—which is syntactically a 1D verbal string (e.g., *"What is the sound of one hand clapping?"* [44])—function as the functional opposite of a verbal knot? We distinguish between:

- **A Selfish Knot ($\mathcal{K}_{\text{selfish}}$):** A self-sealing belief structure that binds the self-model and recruits auxiliary reasoning to perpetuate its own closure [6].
- **A Selfless Knot ($\mathcal{K}_{\text{selfless}}$):** A deliberately encountered paradoxical structure engineered to expose the binding mechanism itself and reveal the meta-knot of the self without requiring its own permanent preservation.

In established scholarship, Gregory Bateson modeled the kōan as a *therapeutic double bind* that compels the practitioner to step outside a locked meta-communicative frame (Logical Type $n+1$) [26, 28], while Victor Sōgen Hori demonstrated that kōans use the *identity of opposites* to exhaust dualistic subject–object categorization (*'khor gsum* [45]) and trigger non-dual insight [44]. Within our candidate predictive-processing model [46, 47, 48], we hypothesize that because a kōan admits zero stable fixed points in 1D propositional graph traversal ($f(x) = \neg x$), sustained attentional precision on the paradox exhausts top-down auxiliary generation and induces a **precision down-weighting ($\Delta\Pi_L < 0$) on the apex self-loop**, shifting the self-model from transparency toward opacity ($\tau \to 0$) [18].

4.3 Wordless Visualization: Medium-Relative vs. Representational Freedom

A complementary everyday pathway for escaping verbal knots is **wordless visualization**: deliberately shifting cognition from 1D verbal loops into silent 2D–4D visuospatial manifolds. Our framework

decomposes this shift into three distinct functional operations:

Definition 4.1 (The Triadic Operations of Cross-Modal Shift)**.** Transitioning from a knotted verbal loop $\mathcal{L}_{\text{verbal}}$ to a silent visuospatial manifold $\mathcal{V}_{\text{spatial}}$ involves up to three operations:

1. **Displacement (Attentional Decoupling):** Ceasing to feed working-memory rehearsal into phonological loops by redirecting attention to visuospatial sketchpads [36].
2. **Reorganization (Topological Re-Mapping):** Utilizing continuous spatial geometry to lay out relational dependencies synchronously, exposing hidden circularities or missing variables.
3. **Disidentification (Metacognitive Opacity / Decentering):** Recognizing through the act of cross-modal translation that **even the new visual or spatial manifold is merely a constructed representation, not reality itself** ($\tau \to 0$) [49, 18].

This triad establishes our most important analytical boundary:

- **Medium-Relative Freedom (Displacement + Reorganization):** Escaping the specific syntactic traps of 1D verbal chains by shifting into a spatial or metaphorical manifold. Without *Disidentification*, however, an intelligent thinker can fall prey to **cross-modal knot migration**—reifying a geometric diagram or multidimensional model with the same dogmatic certainty previously attached to words.
- **Representational Freedom (Disidentification / Opacity $\tau \to 0$):** The metacognitive capacity to translate fluidly across verbal, metaphorical, and spatial representations without mistaking *any* representational frame for ontological ground. In clinical psychology, this corresponds to **decentering**—comprising *meta-awareness*, *disidentification from internal experience*, and *reduced reactivity to thought content* (Bernstein, Hadash, & Fresco, 2015, 2019 [49, 50, 51])—and **cognitive defusion** (Hayes & Masuda, 2004 [53]).

5 A Preregisterable Empirical Research Programme

Because existing measures of decentering [49] and dual-coding [36] do not by themselves establish the proposed Meta-Knot mechanisms, we outline a preregisterable experimental programme designed to test whether metaphorical reframing, wordless visualization, and explicit disidentification independently and synergistically predict **three distinct outcomes**:

1. **Outcome 1 — Representation Change:** The ability to accurately re-express an argument's causal structure in a novel conceptual or visuospatial format.
2. **Outcome 2 — Epistemic Revision:** Normative belief updating in response to disconfirming empirical evidence on identity-salient topics.
3. **Outcome 3 — Decentering:** Reduced cognitive fusion, reduced autonomic defensiveness, and heightened disidentification from self-relevant thoughts [49, 51, 53].

5.1 Experimental Protocol 1: Testing Cross-Modal Reframing and Disidentification

Participants stratified by cognitive ability (g / numeracy / CRT [11, 15]) and baseline conviction on contested topics are randomized across Conditions C1–C4 (Table 4) prior to evaluating disconfirming covariance data [13, 15]. Preregistered hypotheses test whether C3 (*Wordless Spatial Mapping*) produces high *Representation Change* (Outcome 1) but only partial *Epistemic Revision* (Outcome 2) once ideological labels are unmasked (*knot migration*), whereas C4 (*Spatial Mapping + Disidentification*) uniquely maximizes durable *Epistemic Revision* (Outcome 2) mediated by state *Decentering* (Outcome 3) [49].

Table 4: Preregisterable Experimental Matrix Dissociating Representation Change, Epistemic Revision, and Decentering

Experimental Condition	**Active Mechanisms**	**Predicted Outcome 1: Representation Change**	**Predicted Outcome 2: Epistemic Revision**	**Predicted Outcome 3: Decentering ([49])**
C1. Serial Verbal Argumentation (1D Graph Control)	Phonological / deductive rehearsal.	Low (remains confined to initial lexical graph).	Low baseline updating under identity threat ([11]).	Low (high fusion $\tau \to 1$; elevated autonomic threat response).
C2. Metaphorical Cross-Domain Blending	Polysemous domain mapping ([39]).	Moderate–High (maps structure onto isomorphic domain).	Moderate updating while labels are masked; vulnerable to auxiliary re-rationalization.	Low–Moderate (relaxes lexical rigidity without explicit opacity).
C3. Wordless Spatial Manifold Mapping	*Displacement* + *Reorganization* ([36]).	High (synchronous topological representation of causal flows).	Moderate (*medium-relative freedom*); susceptible to *visual knot migration* when identity cues return.	Low–Moderate (shifts medium without targeting self-model fusion).
C4. Spatial Mapping + Disidentification Induction	*Displacement* + *Reorganization* + *Disidentification*.	High (cross-modal structural reorganization).	**Highest durable belief revision** across both masked and unmasked identity trials.	**High** (elevated state decentering [49] and cognitive defusion [53]).

5.2 Experimental Protocol 2: Auditing Argument Topology in Deductive Corpora

To empirically evaluate whether complex deductive texts (such as existential-risk arguments [9] or ideological manifestos [6]) exhibit *Shared Representational Capture*, independent raters and computational graph-extraction models map the text across three preregistered metrics: (1) **Lexical Drift Distance** (cosine distance of contextual embeddings for core terms from premise to conclusion), (2) **Cycle & Ratchet Index** (ratio of unidirectional forward inferences to falsifiable back-propagation links in the extracted causal graph), and (3) **Cross-Modal Invariance** (whether domain experts reach the same probability estimates when the prose argument is translated into a label-free stock-and-flow manifold simulation).

6 Conclusions

Beginning from the recognition that *there are no absolute truths to be found in words alone*, this monograph has synthesized formal limitative theorems [1, 2, 4], dual-coding and conceptual blending research [36, 39], epistemology of self-sealing belief systems [6], and contemplative science [18, 46, 49] into a unified theoretical and empirical architecture. When cognition is restricted to 1D symbolic graph traversal, semantic drift, axiomatic stacking, reflexive inoculation, and inferential ratcheting can lock even highly intelligent reasoners into self-sealing *selfish knots* that preserve subjective certainty while correspondence with reality deteriorates [6, 11]. Lifting problems into metaphorical or multidimensional spatial manifolds provides *medium-relative freedom* via *Displacement* and *Reorganization*. Yet enduring *representational freedom* requires *Disidentification*: rendering the meta-knot of the self-model phenomenally opaque ($\tau \to 0$) so that awareness can use every representational medium—verbal, mathematical, or visual—without mistaking the map for the territory.

References

[1] Gödel, K. (1931/2026). On Formally Undecidable Propositions of *Principia Mathematica* and Related Systems (*Gödel's Incompleteness Theorems*). Archival synthesis available at Wikipedia: Gödel's Incompleteness Theorems.

[2] Tarski, A. (1936/2026). The Concept of Truth in Formalized Languages (*Tarski's Undefinability Theorem*). Archival synthesis available at Wikipedia: Tarski's Undefinability Theorem.

[3] StackExchange Philosophy Logic Symposium (2011). Formal Scope of Tarski's Undefinability Theorem versus Gödel's Incompleteness Theorems in Formal Languages. Available at Philosophy StackExchange Archive.

[4] Preprints.org Monograph in Logic and Computation (2026). Definability, Compression, and Truth in a Layered Account of Computation: Rice's Theorem (1953), Tarski's Undefinability Theorem, and Internal vs. External Grounding. Available at Preprints Full-Text PDF.

[5] arXiv Theoretical Computer Science Monograph (2026). Rice's Theorem under Self-Modification: Elevation Operators and Undecidability of Semantic Properties in Recursive Systems. Available at https://arxiv.org/html/2609.11326v3.

[6] Boudry, M., & Hofhuis, S. (2024). On Epistemic Black Holes: How Self-Sealing Belief Systems Develop and Evolve. *Theoria*, 90(4), 429–447 (DOI: 10.1111/theo.12554). Full text available at EconStor Full-Text PDF.

[7] Boudry, M., & Hofhuis, S. (2024/2026). On Epistemic Black Holes: How Self-Sealing Belief Systems Develop and Evolve (*Theoria* Bibliographic Record). Available at ResearchGate Publication 364933952.

[8] LessWrong Epistemology Archive (2010). Existential Risk, Deductive Chains, and Public Relations in *The Sequences*. Available at LessWrong Archive.

[9] Christiano, P., & LessWrong Technical Symposium (2022). Where I Agree and Disagree with Eliezer: Evaluating Deductive Doom Arguments vs. Empirical World-Models. Available at LessWrong Technical Analysis.

[10] Stanovich, K. E. (2009). *What Intelligence Tests Miss: The Psychology of Rational Thought*. New Haven, CT: Yale University Press. Available at Yale University Press Monograph Archive.

[11] Stanovich, K. E., West, R. F., & Toplak, M. E. (2013). Myside Bias, Rational Thinking, and Intelligence. *Current Directions in Psychological Science*, 22(4), 259–264 (SAGE Publications). Available at ResearchGate Publication 258127972.

[12] Stanovich, K. E., West, R. F., & Toplak, M. E. (2013). Myside Bias, Rational Thinking, and Intelligence (Archival Record). Available at ScienceOpen Record.

[13] Kahan, D. M., Peters, E., Dawson, E. C., & Slovic, P. (2017). Motivated Numeracy and Enlightened Self-Government. *Behavioural Public Policy*, 1(1), 54–86. Available at Cambridge Core and University of Michigan RCGD PDF.

[14] Kahan, D. M. (2013). Ideology, Motivated Reasoning, and Cognitive Reflection. *Judgment and Decision Making*, 8(4), 407–424. Available at University of Pennsylvania Annenberg PDF.

[15] Persson, E., Andersson, D., Koppel, L., Västfjäll, D., & Tinghög, G. (2021). A Preregistered Replication of Motivated Numeracy. *Cognition*, 214, 104768. Available at PubMed 34051421.

[16] Persson, E., Andersson, D., Koppel, L., Västfjäll, D., & Tinghög, G. (2020). Motivated Numeracy and Active Reasoning in a Western European Sample. *Behavioural Public Policy*, 8(1), 24–46. Available at ResearchGate Publication 343451436.

[17] Cambridge Core Behavioural Public Policy (2021). Motivated Reasoning, Fast and Slow (Reviewing Replications of Kahan et al. by Persson et al., 2021). *Behavioural Public Policy*. Available at Cambridge Core Article.

[18] Metzinger, T. (2003). Phenomenal Transparency and Cognitive Self-Reference. *Phenomenology and the Cognitive Sciences*, 2(4), 353–393. Available at ResearchGate Publication 226336928.

[19] Metzinger, T. (2007). Self Models: Phenomenal Transparency and Opacity in the Self-Model Theory of Subjectivity (SMT). *Scholarpedia*, 2(10), 4174. Available at http://www.scholarpedia.org/article/Self_models.

[20] Metzinger, T. (2011). The No-Self Alternative. In S. Gallagher (Ed.), *The Oxford Handbook of the Self* (pp. 279–296). Oxford University Press. Available at Full Text PDF.

[21] The Map of Consciousness Project (2025). Self-Model Theory (PSM) and Anti-Substantialist Theories of the Self. Available at Map of Consciousness Archive.

[22] Metzinger, T. (2009/2026). *The Ego Tunnel: The Science of the Mind and the Myth of the Self* (Analytical Overview). Available at https://sobrief.com/books/the-ego-tunnel.

[23] Masood, A. (2025). Hofstadter's "Strange Loop" of Consciousness: Note on the Self as a Feedback System and Methodological Appraisal in Light of Predictive Coding. Available at Medium Consciousness Monograph.

[24] Laing, R. D. (1970). *Knots*. London: Tavistock Publications / Pantheon Books. Archival excerpts available at Goodreads: R.D. Laing Knots and Knots Work Archive.

[25] Laing, R. D. (1970/2025). Excerpts from *Knots* by R. D. Laing. *Read A Little Poetry*. Available at Read A Little Poetry Archive.

[26] Bateson, G. (1972). *Steps to an Ecology of Mind: Collected Essays in Anthropology, Psychiatry, Evolution, and Epistemology*. San Francisco: Chandler Publishing Company. Full text available at UCSB Orfalea Center PDF.

[27] Bateson, G. (1972). Toward a Theory of Schizophrenia & Logical Types in Double Bind Situations. *Steps to an Ecology of Mind* (UNAM Philosophical Archive). Available at UNAM Archive PDF.

[28] Bateson, G. (1972). Cybernetics of "Self" and Ecology of Mind. Available at Media Studies Asia PDF.

[29] Kauffman, L. H. (1995). Knot Logic and Laws of Form: An Exploration in Mathematics and Foundations. In *Knots and Applications*, World Scientific. Available at UIC Mathematics PDF.

[30] Kauffman, L. H. (2016). Laws of Form and Form Dynamics: Self-Reference and Recursive Forms. Available at ResearchGate Publication 261850139.

[31] Tydecks, R. (2024). Spencer-Brown: *Laws of Form* and Negative Self-Reference in Knot Logic. *Multiple Mathematik*. Available at Multiple Mathematik Archive.

[32] Corballis, M. C. (2014). Left Brain, Right Brain: Facts and Fantasies. *PLOS Biology*, 12(1), e1001767. Available at PLOS Biology and PubMed Central PMC3897366.

[33] Taylor & Francis Symposium on Iain McGilchrist (2019). McGilchrist's Hemispheric Homunculi: Critical Assessment of the Divided Brain Hypothesis. *Religion, Brain & Behavior*. Available at Taylor & Francis Online.

[34] McGilchrist, I. (2009). *The Master and His Emissary: The Divided Brain and the Making of the Western World*. New Haven: Yale University Press. Critical review available at Middle Way Society Analysis.

[35] Christensen, J. M. (2025). Analysis of *The Master and His Emissary* by Iain McGilchrist. Available at Book Summary Archive.

[36] Paivio, A. (2010). Dual Coding Theory and the Mental Lexicon. *The Mental Lexicon*, 5(2), 205–230. Available at ResearchGate Publication 233645196.

[37] Piccardi, L., et al. (2021). Haptic Recognition Memory and Lateralisation for Verbal and Nonverbal Stimuli Under Dual Coding Theory. *Memory*, Taylor & Francis. Available at Taylor & Francis Online.

[38] Anderson, M. L., et al. (2014). Conceptual Mappings and Neural Reuse in Conceptual Metaphor Theory (Lakoff & Johnson) and Blending Theory. *Frontiers in Psychology / PMC*. Available at PubMed Central PMC4009410.

[39] Turner, M., & Fauconnier, G. (2002). Blending and Metaphor. Available at https://mark-turner.org/blendaphor.html.

[40] Fauconnier, G., & Turner, M. (2024). Introduction to Conceptual Blending. Available at ResearchGate Publication 384457898.

[41] Universität Freiburg Germanistik (2023). Metaphor and Counterfactual Thinking: Lakoff & Johnson (1980) and Fauconnier & Turner (2002). *FRAGL 29*. Available at Uni-Freiburg PDF.

[42] IntechOpen (2020). Cognitive Semiotics and Conceptual Blend. Available at https://www.intechopen.com/chapters/72178.

[43] Zenodo Monograph (2026). Conceptual Blending as an Important Theory of Meaning Construction (Lakoff, Johnson, Fauconnier, Turner). Available at https://zenodo.org/records/20025648.

[44] Hori, V. S. (2003). *Zen Sand: The Book of Capping Phrases for Kōan Practice* (Doctrinal Synthesis on Nonduality and the Identity of Opposites in Kōan Study). University of Hawai'i Press. Summary and citations available at Wikipedia: Koan.

[45] Capriles, E. (2007). *Beyond Mind: Beyond Being, Beyond Mind, Beyond History* (Analysis of R.D. Laing, Threefold Thought-Structure *'khor gsum*, and Nondual Liberation). Available at Scribd Monograph Archive.

[46] Laukkonen, R. E., & Slagter, H. A. (2021). From Many to (n)one: Meditation and the Plasticity of the Predictive Mind. *Neuroscience & Biobehavioral Reviews*, 128, 199–217. Available at PubMed 34139248 and Southern Cross University Portal.

[47] Laukkonen, R. E., & Slagter, H. A. (2021). From Many to (n)one: Meditation and the Plasticity of the Predictive Mind (Vrije Universiteit Research Record). Available at VU Research Portal.

[48] Laukkonen, R. E., Sacchet, M. D., Barendregt, H., Devaney, K. J., Chowdhury, A., & Slagter, H. A. (2023). Cessations of Consciousness in Meditation: Advancing a Scientific Understanding of *Nirodha Samāpatti*. *Progress in Brain Research*, 280, 61–87. Available at Harvard MGH Contemplative Neuroscience PDF.

[49] Bernstein, A., Hadash, Y., & Fresco, D. M. (2019). Metacognitive Processes Model of Decentering: Emerging Methods and Insights. *Current Opinion in Psychology*, 28, 245–251. Available at PubMed 30908987 and ResearchGate Publication 329455400.

[50] Bernstein, A., Hadash, Y., & Fresco, D. M. (2019). Metacognitive Processes Model of Decentering (Full-Text Repository). Available at PubMed Central PMC7459352.

[51] Bernstein, A., Hadash, Y., Lichtash, Y., Tanay, G., Shepherd, K., & Fresco, D. M. (2015). Decentering and Related Constructs: A Critical Review and Metacognitive Processes Model. *Perspectives on Psychological Science*, 10(5), 599–617. Available at PubMed 26385999.

[52] EMA Live Glossary (2026). Witness Consciousness, Metacognitive Awareness (Teasdale et al., 2002), and Decentering (Fresco et al., 2007). Available at EMA Glossary Archive.

[53] Masuda, A., Hayes, S. C., Sackett, C. F., & Twohig, M. P. (2004). Cognitive Defusion and Self-Relevant Negative Thoughts: Examining the Impact of a Ninety Year Old Technique. *Behaviour Research and Therapy*, 42(4), 477–485. Available at ResearchGate Publication 7248125.